Aritrash Sarkar

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EDUCATION

Meghnad Saha Institute of Technology — *Expected 2028*
B.Tech in Computer Science and Engineering — *Kolkata, West Bengal*

TECHNICAL SKILLS

Languages: C, Rust, C++, Assembly (x86, ARM), Verilog
Architecture & Core: HW/SW Co-Design, Kernel Development, Chip Microarchitecture, VLSI Design, SoC Power Integrity, Embedded Systems
Hardware & Tooling: QEMU, ESP32, Raspberry Pi 4, Yosys, Icarus Verilog, Bare-Metal Initialization

RESEARCH & PATENTS

Ternary Photonic Architecture (TPA-1) — *Independent Researcher*
Non-binary polarization-coded photonic architecture

- Architecting a 3D stacked die framework featuring MRR technology and a custom Photonic ALU.
- Designing custom assembly language and conducting RTL analysis to optimize execution and bypass traditional CMOS bottlenecks.

Patent-Pending Edge Hardware — *Systems Architect*
Hardware, Sensor Fusion & Edge ML Integration

- **ADEMS:** Engineered an Active Decentralized Embedded Management System integrating custom hardware and Convolutional Neural Networks across ESP32 and Raspberry Pi nodes.
- **Desktop AQI Sensor:** Designed physical particle size detection hardware via Laser Mie Scattering (PD+TIA detection) interfaced with an edge-deployed TinyML model.

ARCHITECTURE & LOW-LEVEL PROJECTS

Obisync [GitHub] — *Systems Architect*
Bare-Metal ARM OS Kernel (Rust)

- Architected and successfully booted a custom Rust-based kernel on physical Raspberry Pi 4 hardware, bypassing virtualization to handle real-world bare-metal initialization.

Aether WebOS [GitHub] — *Lead Developer*
ARM-Native Headless Server OS Kernel (C)

- Engineered a custom headless server kernel featuring minimal drivers and VirtIO support on QEMU (Copyright application submitted).

Experimental Kernels: Aether Grim & EdgeCloud [GitHub] — *Lead Developer*
x86 & ARM64 Microarchitecture (Rust)

- Developed **Aether Grim**, an x86 bare-metal kernel, and prototyped **Aether EdgeCloud**, an ARM64 EL2 micro-hypervisor.

VeriWaveX [GitHub] — *Software Engineer*
Open-Source Verilog IDE

- Engineered a Verilog IDE (v2.0.0) integrating Icarus Verilog and GTKWave, featuring native Yosys synthesis support for Windows environments to streamline local hardware design workflows.

CERTIFICATIONS

Onchip Power Reliability under Semiconductors | *Ansys Innovation Hub* — Jun 2026
Focus: SoC Power Delivery Networks (PDNs) and physical manufacturing constraints.